# Problem Set 3

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Due: 11:59pm Central, Friday 10/03/2025. Submit a PDF to Canvas.

## Instructions

This problem set will be graded on a complete/incomplete basis. Each question is worth 2 points. You will get full credit (2 points) as long as you attempt to answer the question. The questions are designed to help you prepare for the midterm on 10/08.

## Problem 1

Suppose you have a random sample $\{(x_i, y_i)\}$ of size $N$, where each $y_i$ is $1 \times 1$ (a scalar outcome), and each $x_i$ is also $1 \times 1$ (a scalar regressor). If you run a linear regression of $y_i$ on $x_i$ without a constant, show the regression coefficient (OLS estimator) is:

$$\hat{\beta} = \frac{\frac{1}{N}\sum_i y_i x_i}{\frac{1}{N}\sum_i x_i^2}.$$

*Hint:* $\hat{\beta} = \arg\min_b \frac{1}{N}\sum_i (y_i - x_i b)^2$.

**Solution.** We want to solve

$$\hat{\beta} = \arg\min_b Q(b), \quad \text{where } Q(b) = \frac{1}{N}\sum_{i=1}^{N}(y_i - x_i b)^2.$$

Expanding the quadratic:

$$Q(b) = \frac{1}{N}\sum_{i=1}^{N}\left(y_i^2 - 2y_i x_i b + x_i^2 b^2\right).$$

Differentiate $Q(b)$ with respect to $b$:

$$Q'(b) = \frac{1}{N}\sum_{i=1}^{N}\left(-2y_i x_i + 2x_i^2 b\right).$$

Simplify:

$$Q'(b) = -\frac{2}{N}\sum_{i=1}^{N} y_i x_i + \frac{2}{N} b \sum_{i=1}^{N} x_i^2.$$

Set $Q'(b) = 0$ for minimization:

$$-\frac{2}{N}\sum_{i=1}^{N} y_i x_i + \frac{2}{N} b \sum_{i=1}^{N} x_i^2 = 0.$$

Rearrange for $b$:

$$b = \frac{\sum_{i=1}^{N} y_i x_i}{\sum_{i=1}^{N} x_i^2}.$$

Therefore, the OLS estimator is

$$\hat{\beta} = \frac{\frac{1}{N}\sum_i y_i x_i}{\frac{1}{N}\sum_i x_i^2}.$$

□

Note that, since $\mathrm{Var}(x_i) > 0$, the denominator $\sum_i x_i^2 > 0$, ensuring $\hat{\beta}$ is well-defined.

# Problem 2

Prove the Frisch–Waugh theorem for the sample OLS regression coefficients by showing that the $j$th row of $\hat{\beta}$ is

$$\hat{\beta}_j = \left[\frac{1}{N}\sum_{i=1}^{N} \hat{\xi}_i^2\right]^{-1} \left[\frac{1}{N}\sum_{i=1}^{N} \hat{\xi}_i y_i\right],$$

where $\hat{\xi}_i$ is the residual from an auxiliary OLS regression of $x_{ji}$ on all the other $x$'s:

$$x_{ji} = x'_{(\sim j)i}\hat{\pi} + \hat{\xi}_i.$$

*Hint:* review lecture 4.

**Solution.** The OLS estimator from the full model satisfies the normal equations

$$X'(y - X\hat{\beta}) = 0 \iff \begin{bmatrix} x_j' x_j & x_j' X_{-j} \\ X_{-j}' x_j & X_{-j}' X_{-j} \end{bmatrix} \begin{bmatrix} \hat{\beta}_j \\ \hat{\beta}_{-j} \end{bmatrix} = \begin{bmatrix} x_j' y \\ X_{-j}' y \end{bmatrix}. \tag{NE}$$

Let $P_{-j} = X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'$ be the projection matrix onto the column space of $X_{-j}$ and $M_{-j} = I - P_{-j}$ the residual-maker matrix. Premultiplying the model by $M_{-j}$ yields

$$M_{-j}y = M_{-j}x_j\,\beta_j + M_{-j}u \qquad \text{since } M_{-j}X_{-j} = 0.$$

Because $M_{-j}$ is symmetric and idempotent, the OLS coefficient from this transformed regression is

$$\hat{\beta}_j = \frac{(M_{-j}x_j)'(M_{-j}y)}{(M_{-j}x_j)'(M_{-j}x_j)} = \frac{x_j' M_{-j} y}{x_j' M_{-j} x_j}. \tag{1}$$

Now consider the auxiliary regression of $x_j$ on $X_{-j}$:

$$x_j = X_{-j}\hat{\pi} + \hat{\xi} \quad \Longrightarrow \quad \hat{\xi} = M_{-j}x_j,$$

so $\hat{\xi}$ is exactly the vector of residuals from projecting $x_j$ on *all* the other covariates. Because $M_{-j}$ is symmetric,

$$x_j' M_{-j} y = (M_{-j}x_j)'y = \hat{\xi}'y \qquad \text{and} \qquad x_j' M_{-j} x_j = (M_{-j}x_j)'(M_{-j}x_j) = \hat{\xi}'\hat{\xi}.$$

Substituting these into (1) gives

$$\hat{\beta}_j = \frac{\hat{\xi}'y}{\hat{\xi}'\hat{\xi}} = \left[\tfrac{1}{N}\hat{\xi}'\hat{\xi}\right]^{-1}\left[\tfrac{1}{N}\hat{\xi}'y\right] = \left[\tfrac{1}{N}\sum_{i=1}^{N}\hat{\xi}_i^2\right]^{-1}\left[\tfrac{1}{N}\sum_{i=1}^{N}\hat{\xi}_i\, y_i\right],$$

which is the desired sample Frisch–Waugh representation.

□

# Problem 3

We observe an outcome variable $y_i$ and covariates $x_i$ and $z_i$ in a sample of $N$ observations. Assume that the sample means of $x_i$ and $z_i$ are both 0 (i.e., $\bar{x} = 0$, $\bar{z} = 0$). Consider the regression model:

$$y_i = a_0 + a_1 x_i + v_i \tag{4}$$

which we estimate by OLS, obtaining OLS regression coefficients $\hat{a}_0$ and $\hat{a}_1$. Show that if $x_i$ is *orthogonal* to $z_i$ in the observed sample $\left(\frac{1}{N}\sum_i x_i z_i = 0\right)$, then when you estimate the regression model:

$$y_i = b_0 + b_1 x_i + b_2 z_i + e_i \tag{5}$$

by OLS, you will get

$$\hat{b}_0 = \hat{a}_0, \quad \text{and} \quad \hat{b}_1 = \hat{a}_1.$$

*HINT:* Write out the three FOC's for the OLS coefficients from model (5). Compare these with the two FOC's from model (4) and use the information you are given.

**Solution.** To my understanding, this problem is designed to help us reason through and formally show why adding an *orthogonal* covariate $z_i$ to a regression does not change the OLS estimates of the coefficients on the other covariates. In other words, omitting a covariate that is orthogonal to the included regressors does *not* lead to omitted variable bias. The goal is to prove that when $x_i$ and $z_i$ are orthogonal in the sample ($\frac{1}{N}\sum_i x_i z_i = 0$), the estimated coefficients on $x_i$ and the intercept remain unchanged when $z_i$ is added to the regression model. This demonstrates that the inclusion or exclusion of $z_i$ is inconsequential for estimating the effect of $x_i$ on $y_i$ in this special case.

Conceptually, this problem distinguishes between the "short" and "long" regression models:

- The **short regression** omits $z_i$ and estimates $y_i = a_0 + a_1 x_i + v_i$.
- The **long regression** includes $z_i$ and estimates $y_i = b_0 + b_1 x_i + b_2 z_i + e_i$.

Whether the coefficients from the short and long regressions differ depends entirely on whether the omitted variable $z_i$ is correlated with the included variable $x_i$. When $z_i$ is orthogonal to $x_i$, the two models yield the same estimated coefficients on $x_i$ and the same intercept term.

We are given $\bar{x} = \bar{z} = 0$ and sample orthogonality $\frac{1}{N}\sum_i x_i z_i = 0$. The short model is $y_i = a_0 + a_1 x_i + v_i$ and the long model is $y_i = b_0 + b_1 x_i + b_2 z_i + e_i$.

For the FOC of the "short" model, we define the objective function as follows:

$$Q_S(a_0, a_1) = \frac{1}{N}\sum_{i=1}^{N}(y_i - a_0 - a_1 x_i)^2.$$

From this we get the following FOCs:

$$\frac{\partial Q_S}{\partial a_0} = \frac{2}{N}\sum_i (a_0 + a_1 x_i - y_i) = 0, \qquad \frac{\partial Q_S}{\partial a_1} = \frac{2}{N}\sum_i x_i(a_0 + a_1 x_i - y_i) = 0.$$

We then solve the first FOC:

$$\begin{aligned}
\frac{2}{N}\sum_i (a_0 + a_1 x_i - y_i) = 0 &\Longrightarrow a_0\Big(\tfrac{1}{N}\sum_i 1\Big) + a_1\Big(\tfrac{1}{N}\sum_i x_i\Big) - \Big(\tfrac{1}{N}\sum_i y_i\Big) = 0 \\
&\Longrightarrow a_0 + a_1\bar{x} - \bar{y} = 0 \\
&\Longrightarrow \boxed{\hat{a}_0 = \bar{y}} \quad (\text{since } \bar{x} = 0).
\end{aligned}$$

We then substitute $\hat{a}_0$ into the second FOC:

$$\begin{aligned}
\frac{2}{N}\sum_i x_i(\hat{a}_0 + \hat{a}_1 x_i - y_i) = 0 &\Longrightarrow \underbrace{\Big(\tfrac{1}{N}\sum_i x_i\Big)}_{=\bar{x}=0}\hat{a}_0 + \hat{a}_1\Big(\tfrac{1}{N}\sum_i x_i^2\Big) - \Big(\tfrac{1}{N}\sum_i x_i y_i\Big) = 0 \\
&\Longrightarrow \boxed{\hat{a}_1 = \frac{\frac{1}{N}\sum_i x_i y_i}{\frac{1}{N}\sum_i x_i^2} = \frac{\widehat{Cov}(x_i, y_i)}{\widehat{Var}(x_i)}}.
\end{aligned}$$

We also do the same for the "long" regression and find the FOCs and check what they imply. To do this we define first the objective function:

$$Q_L(b_0, b_1, b_2) = \frac{1}{N}\sum_{i=1}^{N}(y_i - b_0 - b_1 x_i - b_2 z_i)^2.$$

From this we can derive the FOCs:

$$\frac{\partial Q_L}{\partial b_0} = \frac{2}{N}\sum_i (b_0 + b_1 x_i + b_2 z_i - y_i) = 0, \quad \frac{\partial Q_L}{\partial b_1} = \frac{2}{N}\sum_i x_i(b_0 + b_1 x_i + b_2 z_i - y_i) = 0,$$

$$\frac{\partial Q_L}{\partial b_2} = \frac{2}{N}\sum_i z_i(b_0 + b_1 x_i + b_2 z_i - y_i) = 0.$$

The intercept FOC gives

$$\begin{aligned}
\frac{2}{N}\sum_i (b_0 + b_1 x_i + b_2 z_i - y_i) = 0 &\Longrightarrow b_0 + b_1\bar{x} + b_2\bar{z} - \bar{y} = 0 \\
&\Longrightarrow \boxed{\hat{b}_0 = \bar{y}}
\end{aligned}$$

This is true considering that we are given that we are to assume that $\bar{x} = 0$ and $\bar{z} = 0$. We can then move on to find $\hat{b}_1$; the slope-on-$x$ FOC can be solved as

$$\frac{2}{N}\sum_i x_i(\hat{b}_0 + \hat{b}_1 x_i + \hat{b}_2 z_i - y_i) = 0 \Longrightarrow \hat{b}_1\Big(\tfrac{1}{N}\sum_i x_i^2\Big) + \hat{b}_2\Big(\tfrac{1}{N}\sum_i x_i z_i\Big) - \Big(\tfrac{1}{N}\sum_i x_i y_i\Big) = 0$$
$$\Longrightarrow \hat{b}_1 = \frac{\frac{1}{N}\sum_i x_i y_i - \hat{b}_2 \frac{1}{N}\sum_i x_i z_i}{\frac{1}{N}\sum_i x_i^2}.$$

Using the *sample orthogonality* assumption $\frac{1}{N}\sum_i x_i z_i = 0$,

$$\boxed{\hat{b}_1 = \frac{\frac{1}{N}\sum_i x_i y_i}{\frac{1}{N}\sum_i x_i^2} = \hat{a}_1}.$$

Thus $\hat{b}_0 = \hat{a}_0$ and $\hat{b}_1 = \hat{a}_1$ already follow from the FOCs.

This result/method made more intuitive sense to me, but that was because I was scared to use Frisch-Waugh (largely because my understanding of the theorem was muddy at best). I realize now that these types of questions where we have a model with additional controls yet seek to analyze a single coefficient, such as the coefficient on $x_i$ here, is best done when we "partial-out" the other covariates to then get a "more causal" coefficient from the univariate regression of $y_i$ on $x_i$. After hitting my head against the wall with my peers for several hours, I realize now we arrive at the same conclusion via FW when we form an auxiliary regression of $x_i$ on $(1, z_i)$. (*Also forgive me if I used FW or FWL interchangeably in what follows; I mean both to denote Frisch-Waugh–I am tired*). To begin with, FW says: the coefficient on $x_i$ in $y \sim (1, x_i, z_i)$ equals the coefficient from regressing $y_i$ on the *residual* of $x$ after projecting $x_i$ on $(1, z_i)$. We run the auxiliary regression in sample

$$x_i = \pi_0 + \pi_1 z_i + \tilde{\xi}_i,$$

where

$$\frac{1}{N}\sum_i \tilde{\xi}_i = \frac{1}{N}\sum_i \tilde{\xi}_i z_i = 0.$$

From this auxiliary, univariate regression set up, we get the following results from the sample realization of the OLS coefficients:

$$\hat{\pi}_1 = \frac{\frac{1}{N}\sum_i (z_i - \bar{z})(x_i - \bar{x})}{\frac{1}{N}\sum_i (z_i - \bar{z})^2} = \frac{\frac{1}{N}\sum_i x_i z_i}{\frac{1}{N}\sum_i z_i^2} = 0 \quad (\text{by } \bar{x} = \bar{z} = 0 \text{ and } \tfrac{1}{N}\sum_i x_i z_i = 0),$$

and $\hat{\pi}_0 = \bar{x} - \hat{\pi}_1 \bar{z} = \bar{x} = 0$. Therefore the residual equals the original regressor:

$$\hat{\xi}_i := x_i - \hat{\pi}_0 - \hat{\pi}_1 z_i = x_i.$$

FWL (residual-on-original) yields

$$\hat{b}_1 = \frac{\frac{1}{N}\sum_i \hat{\xi}_i\, y_i}{\frac{1}{N}\sum_i \hat{\xi}_i^2} = \frac{\frac{1}{N}\sum_i x_i y_i}{\frac{1}{N}\sum_i x_i^2} = \hat{a}_1,$$

which matches the FOC derivation.

To be honest, I initially approached this question with a sort of "plug-n-chug" mindset, however, upon further review, I realize this has important implications. Since $z_i$ is *orthogonal* to $x$ in the sample, adding (or omitting) $z_i$ leaves the OLS coefficient on $x_i$ (which is $b_1$ in the long model and $a_1$ in the short model) and the intercept ($b_0$ and $a_0$, respectively) unchanged: $\hat{b}_1 = \hat{a}_1$ and $\hat{b}_0 = \hat{a}_0$. Hence omitting an orthogonal covariate does *not* create omitted-variable bias for the coefficient on $x_i$, which, when searching for causality, is great news.

# Problem 4

Suppose you can observe wages, education, immigrant status, and race/ethnicity of workers.

(a) Compare the regressions:

$$\begin{aligned} y_i &= \lambda_0 + \lambda_1 Imm_i + u_i, \qquad \text{(short model)} \\ y_i &= \beta_0 + \beta_1 Imm_i + \beta_2 Educ_i + \epsilon_i \text{ (long model)} \end{aligned}$$

How does $\hat{\lambda}_1$ relate to $\hat{\beta}_1$? Which assumption ensures $\hat{\lambda}_1 > \hat{\beta}_1$? (*Hint:* use omitted variable bias, and define the terms in the formula carefully.)

**Solution. How I'm reading this.** The short model drops education. If education both (i) affects wages and (ii) covaries with immigrant status, then the short slope on *Imm* picks up the direct effect of immigrant status *plus* an education channel. That is a textbook omitted–variable bias (OVB) situation.

**Plan.** Run an auxiliary regression for education on $(1, Imm_i)$, plug it into the long model, and read off how the short-model coefficients $(\hat{\lambda}_0, \hat{\lambda}_1)$ relate to $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)$.

**Auxiliary (first stage, in sample).**

$$Educ_i = \hat{\pi}_0 + \hat{\pi}_1\, Imm_i + \hat{\xi}_i, \qquad \frac{1}{N}\sum_i \hat{\xi}_i = 0, \quad \frac{1}{N}\sum_i \hat{\xi}_i\, Imm_i = 0.$$

(The last two equalities are the OLS residual orthogonality FOCs for the regression of *Educ* on $(1, Imm)$.)

**Plug into the long model (in sample).**

$$\begin{aligned} y_i &= \hat{\beta}_0 + \hat{\beta}_1\, Imm_i + \hat{\beta}_2\, Educ_i + \hat{\varepsilon}_i \\ &= \hat{\beta}_0 + \hat{\beta}_1\, Imm_i + \hat{\beta}_2\Big(\hat{\pi}_0 + \hat{\pi}_1\, Imm_i + \hat{\xi}_i\Big) + \hat{\varepsilon}_i \\ &= \underbrace{\Big(\hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0\Big)}_{=\,\hat{\lambda}_0} + \underbrace{\Big(\hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1\Big)}_{=\,\hat{\lambda}_1}\, Imm_i + \underbrace{\Big(\hat{\beta}_2\hat{\xi}_i + \hat{\varepsilon}_i\Big)}_{=\,\hat{u}_i}. \end{aligned}$$

**Why this matches the short model.** In the short model

$$y_i = \hat{\lambda}_0 + \hat{\lambda}_1\, Imm_i + \hat{u}_i,$$

the OLS FOCs are

$$\frac{1}{N}\sum_i \hat{u}_i = 0 \quad \text{and} \quad \frac{1}{N}\sum_i \hat{u}_i\, Imm_i = 0.$$

With $\hat{u}_i = \hat{\beta}_2\hat{\xi}_i + \hat{\varepsilon}_i$ we have

$$\frac{1}{N}\sum_i \hat{u}_i = \hat{\beta}_2 \underbrace{\left(\tfrac{1}{N}\sum_i \hat{\xi}_i\right)}_{=0} + \underbrace{\left(\tfrac{1}{N}\sum_i \hat{\varepsilon}_i\right)}_{=0} = 0,$$

$$\frac{1}{N}\sum_i \hat{u}_i\, Imm_i = \hat{\beta}_2 \underbrace{\left(\tfrac{1}{N}\sum_i \hat{\xi}_i\, Imm_i\right)}_{=0} + \underbrace{\left(\tfrac{1}{N}\sum_i \hat{\varepsilon}_i\, Imm_i\right)}_{=0} = 0,$$

since by OLS from the long regression $\hat{\varepsilon}_i \perp (1, Imm_i, Educ_i)$ and from the auxiliary regression $\hat{\xi}_i \perp (1, Imm_i)$. Therefore the pair

$$\boxed{\hat{\lambda}_0 = \hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0, \qquad \hat{\lambda}_1 = \hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1}$$

satisfies the short-model FOCs, i.e., these are exactly the short-model OLS coefficients.

**OVB formula and sign.**

$$\boxed{\hat{\lambda}_1 - \hat{\beta}_1 = \underbrace{\hat{\beta}_2}_{\substack{\text{effect of } Educ \\ \text{on } y}} \cdot \underbrace{\hat{\pi}_1}_{\substack{\text{how } Educ \\ \text{varies with } Imm}}}.$$

Hence $\hat{\lambda}_1 > \hat{\beta}_1$ when $\hat{\beta}_2\hat{\pi}_1 > 0$. *Interpretation:* if education raises earnings ($\hat{\beta}_2 > 0$) and immigrants are on average more educated ($\hat{\pi}_1 > 0$), the short regression overstates the immigrant premium. Flip either sign and the bias flips.

*Takeaway.* The short slope equals the long slope plus *bias* = (effect of omitted variable on $y$)$\times$(association of omitted variable with *Imm*).

(b) Split the immigrants into 3 mutually exclusive groups:

- (i) Asian immigrants: workers who answer Asian = 1, Hispanic=0, immigrant =1;
- (ii) Hispanic immigrants: workers who answer Hispanic=1, immigrant=1;
- (iii) Other immigrants: workers who answer Asian=0, Hispanic=0, immigrant=1.

Specify a single regression that allows you to estimate the mean log wage of each immigrant group relative to natives (immigrant=0).

**Solution. How I'll set it up.** I create three dummy variables that identify each immigrant group, where $D_{3i}$ serves as a general "immigrant" indicator and $D_{1i}$ and $D_{2i}$ identify specific subgroups. Natives are the omitted group, so all three dummies are zero for them:

$$D_{1i} = \mathbf{1}\{\text{Asian}_i\}, \qquad D_{2i} = \mathbf{1}\{\text{Hispanic}_i\}, \qquad D_{3i} = \mathbf{1}\{\text{Immigrant}_i\}.$$

By construction, each person falls in exactly one of the following "on/off" combinations:

| Group $g$ | $D_{1i}$ (Asian) | $D_{2i}$ (Hispanic) | $D_{3i}$ (Immigrant) |
|---|---|---|---|
| Native | 0 | 0 | 0 |
| Asian immigrant | 1 | 0 | 1 |
| Hispanic immigrant | 0 | 1 | 1 |
| Other immigrant | 0 | 0 | 1 |

**Regression.** With natives as the omitted (reference) group, I estimate

$$y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i, \qquad u_i \perp (1, D_{1i}, D_{2i}, D_{3i}).$$

**Interpretation of coefficients.** Each coefficient represents an expected log–wage difference conditional on which dummies are turned on:

$$\begin{aligned}
E[y_i \mid D_{1i} = 0, D_{2i} = 0, D_{3i} = 0] &= \beta_0, && \text{(Natives)} \\
E[y_i \mid D_{1i} = 1, D_{2i} = 0, D_{3i} = 1] &= \beta_0 + \beta_1 + \beta_3, && \text{(Asian immigrants)} \\
E[y_i \mid D_{1i} = 0, D_{2i} = 1, D_{3i} = 1] &= \beta_0 + \beta_2 + \beta_3, && \text{(Hispanic immigrants)} \\
E[y_i \mid D_{1i} = 0, D_{2i} = 0, D_{3i} = 1] &= \beta_0 + \beta_3, && \text{(Other immigrants).}
\end{aligned}$$

**What this means.**

- $\beta_0$ is the mean log wage of natives (the baseline group).
- $\beta_3$ measures the average log–wage gap between *immigrants in general* and natives.
- $\beta_1$ and $\beta_2$ measure the additional (within–immigrant) wage differences of Asian and Hispanic immigrants relative to the "other immigrant" group captured by $D_{3i}$.

So in this setup, $\beta_1$, $\beta_2$, and $\beta_3$ together let me recover the mean log wages of each group relative to natives.

**Recovering group gaps relative to natives.** From the expectations above, the mean log–wage gap between each immigrant group and natives can be expressed directly in terms of the estimated coefficients:

$$\begin{aligned}
\text{Asian–Native gap:}\quad & E[y_i \mid D_{1i}=1, D_{2i}=0, D_{3i}=1] - E[y_i \mid D_{1i}=0, D_{2i}=0, D_{3i}=0] \\
& = (\beta_0+\beta_1+\beta_3)-\beta_0 = \beta_1+\beta_3, \\
\text{Hispanic–Native gap:}\quad & E[y_i \mid D_{1i}=0, D_{2i}=1, D_{3i}=1] - E[y_i \mid D_{1i}=0, D_{2i}=0, D_{3i}=0] \\
& = (\beta_0+\beta_2+\beta_3)-\beta_0 = \beta_2+\beta_3, \\
\text{Other–Native gap:}\quad & E[y_i \mid D_{1i}=0, D_{2i}=0, D_{3i}=1] - E[y_i \mid D_{1i}=0, D_{2i}=0, D_{3i}=0] \\
& = (\beta_0+\beta_3)-\beta_0 = \beta_3.
\end{aligned}$$

Hence, each immigrant group's mean wage relative to natives can be recovered as:

$$\begin{aligned}
\text{Asian immigrants: } & \beta_1+\beta_3, \\
\text{Hispanic immigrants: } & \beta_2+\beta_3, \\
\text{Other immigrants: } & \beta_3.
\end{aligned}$$

This makes it easy to interpret $\beta_3$ as the *baseline immigrant–native wage gap*, while $\beta_1$ and $\beta_2$ capture deviations from that baseline among specific immigrant subgroups.

(c) How much of the (log) wage gaps between each immigrant group and native workers can be ex- plained by differences in education attainment? Specify the regression(s) you will need to estimate to answer this question, and provide an answer from your regressions. Hint: consider a pooled regression with education as a control, or separate regressions by immi- grant group. Focus on the "between" component of the decomposition.

**Solution.** So initially I thought it would be necessary to construct a regression of log wage on education for each of the immigrant groups, but because the groups are mutually exclusive (by construction), we can compare *any one* immigrant group to natives, decompose that gap, and then repeat for the other groups. Here I was biased (but not my estimates; +10 points for the pun) and decided to focus on *Hispanic (non-Asian) immigrants* vs. *Natives*. Still, (and as we will see) the same steps apply verbatim to Asians or "Other" immigrants by relabeling and applying the solution to the other two groups accordingly.

So we begin with constructing separate regressions of log wage ($y_i$) on education ($Educ_i$) for Hispanic immigrants and natives such th1at we have:

$$\begin{aligned}
\text{(Hispanics)}\quad & y_i = \beta_0^H + \beta_1^H\, Educ_i + \varepsilon_i^H, \\
\text{(Natives)}\quad & y_i = \beta_0^N + \beta_1^N\, Educ_i + \varepsilon_i^N.
\end{aligned}$$

Let $\bar{y}^H, \bar{y}^N$ be mean log wages and $\overline{Educ}^H, \overline{Educ}^N$ be mean education for each group.

**Algebra for the mean gap.** Using $\bar{y}^g = \beta_0^g + \beta_1^g\, \overline{Educ}^g$ for $g \in \{H, N\}$,

$$\begin{aligned}
\bar{y}^H - \bar{y}^N &= \left(\beta_0^H + \beta_1^H\, \overline{Educ}^H\right) - \left(\beta_0^N + \beta_1^N\, \overline{Educ}^N\right) \\
&= \underbrace{(\beta_0^H - \beta_0^N)}_{\text{unexplained by educ}} + \underbrace{(\beta_1^H - \beta_1^N)\,\overline{Educ}^H}_{\text{within (returns)}} + \underbrace{\beta_1^N\left(\overline{Educ}^H - \overline{Educ}^N\right)}_{\text{between (composition)}}.
\end{aligned}$$

**Sample (what I'd compute).** Estimate the two OLS regressions above and plug in sample means:

$$\widehat{\bar{y}^H - \bar{y}^N} = \underbrace{(\hat{\beta}_0^H - \hat{\beta}_0^N)}_{\text{unexplained by educ}} + \underbrace{(\hat{\beta}_1^H - \hat{\beta}_1^N)\,\overline{Educ}^H}_{\text{within}} + \underbrace{\hat{\beta}_1^N\left(\overline{Educ}^H - \overline{Educ}^N\right)}_{\text{between}}.$$

$$\boxed{\text{Composition (explained by education)} \;=\; \hat{\beta}_1^N\left(\overline{Educ}^H - \overline{Educ}^N\right)}$$

**Why it's OK to do one immigrant group at a time.** This is a *pairwise* Oaxaca–Blinder decomposition with natives as the reference technology (the "returns" $\hat{\beta}_1^N$). For each immigrant group $g \in \{H, \text{Asian}, \text{Other}\}$, the gap splits into:

$$\underbrace{\hat{\beta}_1^N(\overline{Educ}^g - \overline{Educ}^N)}_{\text{between (composition w.r.t. natives)}} + \underbrace{(\hat{\beta}_0^g - \hat{\beta}_0^N) + (\hat{\beta}_1^g - \hat{\beta}_1^N)\,\overline{Educ}^g}_{\text{within (returns/other factors)}}.$$

Because the decomposition is defined for any two groups, I can repeat this calculation for Asians and for "Other" without re-estimating anything jointly; I just swap in that group's regression and mean education.

**Optional pooled alternative (same idea).** I could also run a pooled regression with a Hispanic dummy and *Educ* plus group dummies, or include group$\times$*Educ* interactions to estimate $\beta_1^g$ within one model; the "between" piece is still the native slope times the education mean difference.

# Problem 5

You are tasked to provide a causal analysis of how the inflow of Asian immigrants affects the earnings of native workers. Assume you have a county c-level data $\{y_c, x_c, z_c\}$:

- $y_c$: the change in real log wage among native workers in country $c$ from 1990 to 2000;
- $x_c$: the percent change in the number of immigrants in county $c$ from 1990 to 2000 (if $N_{ct}$ denotes the number of immigrants in country $c$ in year $t$, $x_c = 100 * (N_{c,2000} - N_{c,1990})/N_{c,1990}$.
- $z_c$: the share of restaurants in county $c$ that are owned by Asian immigrants in 1990.

The causal model of interest is:

$$y_c = \beta_1 + \beta_2 x_c + u_c.$$

in which $\beta_2$ is the causal effect of a 1% increase in immigrant population on the change in log wage of native workers, holding all else constant.

(a) Consider an OLS regression of $y_c$ on $x_c$,

$$y_c = \hat{\beta}_1 + \hat{\beta}_2 x_c$$

how would you interpret the coefficient $\hat{\beta}_2$ on $x_c$? If $\text{cov}(x_c, u_c) = 0$ and $E[u_c] = 0$, show that $\hat{\beta}_2$ is a consistent estimator for $\beta_2$ in the causal model of interest ($y_c = \hat{\beta}_1 + \hat{\beta}_2 x_c$).

**Solution.** Given the causal model is

$$y_c = \beta_1 + \beta_2 x_c + u_c,$$

where $x_c$ is the percent change in immigrants and $y_c$ is the change in log wage of natives (county $c = 1, \dots, N$). The coefficient $\beta_2$ is the causal effect on $y_c$ of a one–unit increase in $x_c$ *holding all else constant.* If the exogeneity condition $E[u_c \mid x_c] = 0$ holds (in particular $E[u_c] = 0$ and $cov(x_c, u_c) = 0$), then the OLS slope from regressing $y_c$ on a constant and $x_c$ estimates this causal effect.

With an intercept in the regression, the OLS slope has the sample "covariance over variance" form

$$\hat{\beta}_2 = \frac{\frac{1}{N}\sum_{c=1}^{N}(x_c - \bar{x})(y_c - \bar{y})}{\frac{1}{N}\sum_{c=1}^{N}(x_c - \bar{x})^2} = \frac{\widehat{cov}(x, y)}{\widehat{var}(x)}.$$

We then substitute the model and expand the numerator as follows: $y_c = \beta_1 + \beta_2 x_c + u_c$:

$$\begin{aligned}
\tfrac{1}{N}\sum(x_c - \bar{x})(y_c - \bar{y}) &= \tfrac{1}{N}\sum(x_c - \bar{x})\Big[(\beta_1 + \beta_2 x_c + u_c) - (\beta_1 + \beta_2 \bar{x} + \bar{u})\Big] \\
&= \tfrac{1}{N}\sum(x_c - \bar{x})\Big[\beta_2(x_c - \bar{x}) + (u_c - \bar{u})\Big] \\
&= \beta_2 \underbrace{\tfrac{1}{N}\sum(x_c - \bar{x})^2}_{\widehat{var}(x)} + \underbrace{\tfrac{1}{N}\sum(x_c - \bar{x})(u_c - \bar{u})}_{\widehat{cov}(x,u)}.
\end{aligned}$$

Hence

$$\hat{\beta}_2 = \beta_2 + \frac{\widehat{cov}(x, u)}{\widehat{var}(x)} = \beta_2 + \frac{\frac{1}{N}\sum(x_c - \bar{x})u_c}{\frac{1}{N}\sum(x_c - \bar{x})^2} \quad (\text{since } \tfrac{1}{N}\sum(x_c - \bar{x}) = 0).$$

Now to prove consistency, if $cov(x_c, u_c) = 0$ and $Var(x_c) > 0$, then by the Law of Large Numbers (LLN),

$$\widehat{cov}(x, u) \xrightarrow{p} cov(x, u) = 0, \qquad \widehat{var}(x) \xrightarrow{p} var(x) > 0.$$

Therefore

$$\hat{\beta}_2 = \beta_2 + \frac{\widehat{cov}(x_c, u_c)}{\widehat{var}(x_c)} \xrightarrow{p} \beta_2,$$

so $\hat{\beta}_2$ is a *consistent* estimator of the causal effect $\beta_2$.

*Note.* The $E[u_c] = 0$ piece is automatically absorbed by the intercept. Even if $E[u_c] \neq 0$, writing $\tilde{u}_c = u_c - E[u_c]$ keeps the same slope model $y_c = (\beta_1 + E[u_c]) + \beta_2 x_c + \tilde{u}_c$ with $E[\tilde{u}_c] = 0$. What matters for consistency is $Cov(x_c, u_c) = 0$.

(b) If $E[u_c x_c] \neq 0$ in model (6), we may instrument the endogenous $x_c$. What assumptions must hold for $z_c$ (share of Asian-owned restaurants in 1990) to be a valid instrument for $x_c$?

**Solution. How I'm thinking about it (intuition).** Counties with a larger pre-existing footprint of Asian-owned restaurants in 1990 plausibly have stronger immigrant networks and business infrastructure. That kind of "beachhead" can make later inflows easier (jobs, housing leads, community ties), so it's reasonable that $z_c$ helps predict the *change* in immigrant population, $x_c$. At the same time, $z_c$ should not change native wage growth *except* through its effect on immigrant inflows—once we hold fixed other county fundamentals (e.g., baseline industry mix or long-run trends).

**Formal IV requirements for $z_c$ to be valid for $x_c$.**

(i) *Relevance (first stage):* $z_c$ must predict $x_c$,

$$Cov(x_c, z_c) \neq 0 \quad \Longleftrightarrow \quad x_c = \pi_0 + \pi_1 z_c + \nu_c \ \text{ with } \ \pi_1 \neq 0.$$

(We check this in data via a strong first stage.)

(ii) *Exclusion / Exogeneity:* $z_c$ affects $y_c$ only through $x_c$ and is orthogonal to the structural error,

$$E[z_c\, u_c] = 0.$$

Substantively: conditional on baseline county characteristics, the 1990 share of Asian-owned restaurants should not shift native wage growth directly; any impact of $z_c$ on $y_c$ operates through its effect on immigrant inflows $x_c$.

*Bottom line.* If (i) $z_c$ is a strong predictor of $x_c$ and (ii) $z_c$ is excluded from the native-wage equation (uncorrelated with $u_c$), then $z_c$ is a valid instrument for $x_c$.

(c) Suppose the assumptions for $z_c$ to be a valid instrument for $x_c$ are satisfied. Consider the first-stage regression in the population

$$x_c = \pi_0 + \pi_1 z_c + \nu_i.$$

and the reduced-form regression (population)

$$y_c = \delta_0 + \delta_1 z_c + \epsilon_i.$$

Show:

$$\frac{\delta_1}{\pi_1} = \beta_2$$

**Solution. How I'm thinking about this.** I'm told to assume the IV assumptions for $z_c$ hold and to look at two *population* regressions:

- *First stage (population):* $x_c = \pi_0 + \pi_1 z_c + \nu_c$.
- *Reduced form (population):* $y_c = \delta_0 + \delta_1 z_c + \epsilon_c$.

The structural (causal) model I care about is

$$y_c = \beta_1 + \beta_2 x_c + u_c,$$

where $\beta_2$ is the causal effect of $x_c$ on $y_c$. The *reduced form* is simply the population regression of the outcome on the instrument (and a constant). Econometrically, it captures the *total* effect of $z_c$ on $y_c$, combining the pathway $z_c \to x_c$ (first stage) with the causal effect $x_c \to y_c$ (structural). If $z_c$ only moves $y_c$ by first moving $x_c$ (exclusion), then the RF slope must be the product of those two links.

**Derivation (substitute first stage into the structural model).** Using $x_c = \pi_0 + \pi_1 z_c + \nu_c$ in $y_c = \beta_1 + \beta_2 x_c + u_c$ gives

$$\begin{aligned} y_c &= \beta_1 + \beta_2(\pi_0 + \pi_1 z_c + \nu_c) + u_c \\ &= \underbrace{(\beta_1 + \beta_2\pi_0)}_{=:\ \delta_0} + \underbrace{(\beta_2\pi_1)}_{=:\ \delta_1} z_c + \underbrace{(\beta_2\nu_c + u_c)}_{=:\ \epsilon_c}. \end{aligned}$$

**Why this is the reduced form.** Under the standard IV assumptions

- (relevance) $\pi_1 \neq 0$,
- (exogeneity) $E[z_c u_c] = 0$,
- (first-stage OLS conditions) $E[\nu_c] = 0,\ E[z_c\nu_c] = 0$,

the composite error $\epsilon_c = \beta_2\nu_c + u_c$ is mean zero and orthogonal to $z_c$:

$$E[\epsilon_c] = \beta_2 E[\nu_c] + E[u_c] = 0, \qquad E[z_c\epsilon_c] = \beta_2 E[z_c\nu_c] + E[z_c u_c] = 0.$$

Those are exactly the population normal equations for the regression of $y_c$ on $(1, z_c)$, so the representation above is the *population reduced form*. Consequently its slope equals the population OLS formula:

$$\delta_1 \ = \ \frac{Cov(z_c, y_c)}{Var(z_c)}.$$

Using $y_c = \beta_1 + \beta_2(\pi_0 + \pi_1 z_c + \nu_c) + u_c$,

$$\begin{aligned} Cov(z_c, y_c) &= Cov\Big(z_c,\ \beta_1 + \beta_2\pi_0 + \beta_2\pi_1 z_c + \beta_2\nu_c + u_c\Big) \\ &= \beta_2\pi_1\, Var(z_c) \ + \ \beta_2\, Cov(z_c, \nu_c) \ + \ Cov(z_c, u_c) \\ &= \beta_2\pi_1\, Var(z_c) \qquad (\text{by } E[z_c\nu_c] = E[z_c u_c] = 0). \end{aligned}$$

Therefore

$$\delta_1 \ = \ \frac{Cov(z_c, y_c)}{Var(z_c)} \ = \ \beta_2\pi_1.$$

Provided the instrument is *relevant* ($\pi_1 \neq 0$),

$$\boxed{\frac{\delta_1}{\pi_1} = \beta_2}$$

i.e., the causal effect equals the reduced–form effect of $z_c$ on $y_c$ divided by the first–stage effect of $z_c$ on $x_c$. (As a sanity check: if $\pi_1 = 0$, the ratio is undefined—this is the weak/irrelevant instrument case.)

(d) Do you think $z_c$ is a good instrument? Why or why not? *(Hint: this is an open-ended question. You can argue either way).*

**Solution.** **How I see it.** The causal model we are working with is

$$y_c = \beta_1 + \beta_2 x_c + u_c,$$

where $y_c$ is the change in real log wage among native workers from 1990–2000, $x_c$ is the percent change in immigrant population over that period, and $z_c$ is the share of restaurants in county $c$ owned by Asian immigrants in 1990. The question is whether $z_c$ provides a valid and credible instrument for $x_c$.

**Case 1: $z_c$ may be a good instrument.** There's a strong intuitive argument for *relevance*: counties with a larger pre-existing Asian immigrant presence—proxied by restaurant ownership—are likely to attract more immigrants later. Restaurant ownership signals a stable, established network, which can lower the costs of migration and settlement for new arrivals (family or community members). If these early establishments formed for largely historical or exogenous reasons (say, pre-existing supply chains, or random settlement patterns from earlier decades), then $z_c$ might also satisfy the *exclusion restriction*: the restaurant share in 1990 affects native wage changes only through its influence on subsequent immigrant inflows, not directly through the local labor market.

**Case 2: $z_c$ may not be a good instrument.** The main concern lies with the exclusion restriction. A higher share of Asian-owned restaurants in 1990 may reflect broader economic activity or local demand shocks that are also correlated with native wage growth. For example, restaurant density may proxy for tourism, consumer spending, or urban development—factors that independently raise native wages. In that case, $E[z_c u_c] \neq 0$, since $z_c$ would be correlated with the unobserved determinants of wage growth. In addition, $z_c$ might even relate to the local service economy's structure, meaning it could affect wages directly if restaurants employ a large number of natives. Even though it's unlikely that $z_c$ is completely irrelevant for $x_c$, its possible correlation with unobserved county-level economic growth makes the exclusion restriction hard to defend in practice.

**Where I land.** The first-stage (relevance) condition seems plausible—more restaurants do signal a larger immigrant base—but the exclusion restriction is much trickier. If the restaurant share captures unobserved local prosperity rather than purely migration networks, then $z_c$ would fail as an instrument. This ambiguity makes $z_c$ an *arguably plausible but fragile* instrument: the story makes sense theoretically, but the identifying assumption requires careful justification and robustness checks in any real empirical setting.